

Accepted Manuscript

A comprehensive framework for cloud computing migration using
Meta-synthesis approach

Hamid reza Bazi , Alireza Hasanzadeh , Ali Moeini

PII: S0164-1212(17)30045-6
DOI: [10.1016/j.jss.2017.02.049](https://doi.org/10.1016/j.jss.2017.02.049)
Reference: JSS 9932



To appear in: *The Journal of Systems & Software*

Received date: 31 August 2016
Revised date: 29 January 2017
Accepted date: 24 February 2017

Please cite this article as: Hamid reza Bazi , Alireza Hasanzadeh , Ali Moeini , A comprehensive framework for cloud computing migration using Meta-synthesis approach, *The Journal of Systems & Software* (2017), doi: [10.1016/j.jss.2017.02.049](https://doi.org/10.1016/j.jss.2017.02.049)

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Highlights

- Presenting a comprehensive cloud migration framework.
- Extracting concepts from previous researches and classify them to categories
- Proposing a maturity model to improve migration process

ACCEPTED MANUSCRIPT

A comprehensive framework for cloud computing migration using Meta-synthesis approach

Hamid reza Bazi¹, Alireza Hasanzadeh^{2*}, Ali Moeini³

Abstract

Migration to the cloud computing environment is a strategic organizational decision. Using a reliable framework for migration ensures managers to mitigate risks in the cloud computing technology. Therefore, organizations always search for cloud migration frameworks with dynamic nature as well as integrity beside their simplicity. In previous studies, these important features have received less attention and have not been achieved in an integrated and comprehensive way. The aim of this study is to use a meta-synthesis method for the first time for analysis and synthesis of previous published studies and suggests a comprehensive cloud migration framework. We review more than 657 papers from relevant journals and conference proceedings. The concepts which are extracted from these papers are classified to related sub-categories and categories. Then, our proposed framework based on these concepts and categories is developed. It includes seven main phases (categories) and fifteen sub-categories. To improve the migration process a maturity model called "CIM3" is introduced. Finally, proposed framework and maturity model is evaluated by forming different focus group meetings and taking advantages of the cloud experts' opinion. The results of this research can help managers have a safe and effective migration to cloud computing environment.

Keywords: Cloud Computing, Migration Framework, Meta-Synthesis, Process Maturity Model.

1. Introduction

Cloud computing denotes hosted online services (Markovic et al., 2013). According to American National Institute of Standards and Technology (NIST) definition, "cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction" (Mell and Grance, 2011). Cloud computing provides new capabilities in line with organizational requirements without the need to invest in infrastructure, new human resources and licenses (Siclován, 2012). Cloud computing is a new technology that helps organizations to develop and create competitive advantage with inherent characteristics such as agility, flexibility, scalability, simple and secure accessibility from anywhere, increasing reliability, enhancing fault tolerance and economic efficiency. NIST and International Organization for Standardization (ISO/IEC 17788), have described items such as on-demand self-service, broad network access, resource pooling, rapid elasticity, measured services and multi-tenancy as the main characteristics of cloud computing (Alsaeed and Saleh, 2015). Service models which are offered by cloud computing include, software as a system (SaaS), platform as a service (PaaS) and infrastructure as a service (IaaS) (Mell and Grance, 2011; Rimal et al., 2009; Sharma and Banga, 2013; Tayal, 2011). To deploy cloud computing, organizations can choose one of the deployment models such as private

¹ Department of Information Technology Management, Tarbiat Modares University, Tehran, Iran. Email: hr.bazi@modares.ac.ir

² (*Corresponding author), Department of Information Technology Management, Tarbiat Modares University (TMU), Jalale Alahmad Avenue, P.O. Box 14155-111, Tehran, Iran. Fax: +98-21-82883637. Email: ar_hassanzadeh@modares.ac.ir.

³ Department of Information Technology Management, Tehran University, Tehran, Iran. Email: A.moeini@ut.ac.ir

cloud, community cloud, public cloud and hybrid cloud (as shown in Fig.1) (Adrees et al., 2015; Yaghmaei and Binesh, 2015).

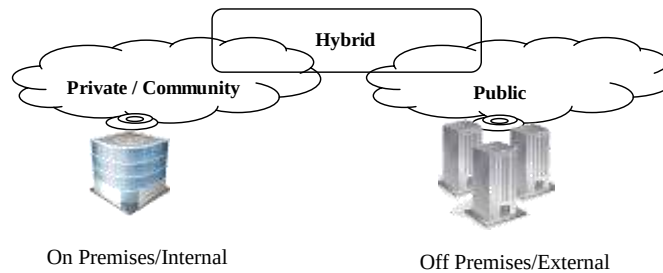


Fig.1. Cloud computing deployment models.

To gain the advantages of cloud computing technology, it is necessary for organizations to migrate their legacy applications and services to cloud environment (Babar and Chauhan, 2011). Migration to cloud is an organizational strategic decision. The aim of various research performed in this field, is to enable organizations to migrate effectively and efficiently from their existing services to cloud solutions. Different organizations should invest in new technologies such as cloud computing to focus on the core business activities and their improvement. An important point in cloud adoption and migration is the mentality of organizations; they must focus on the services instead of infrastructure ownership (Rewatkar and Lanjewar, 2010).

Regarding to the diversity of cloud offerings with various options (such as location, sales model, reproducibility, etc.) for usage, it is difficult to select the most appropriate configuration for infrastructure, best service, and provider (García-Galán et al., 2016). So a framework is required to help the decision makers and perform successful migration. In most studies, Decision Support System (DSS) are designed for migration to the cloud which mostly focused on selecting a cloud service provider. In fact, they considered the cost as a major factor but they ignored comprehensive researching to find the key factors involved in the migration. In addition to choose a provider, organizations need to increase their knowledge and be aware of capabilities, principles and rules of the cloud technology, (Alkhalil et al., 2014). As Jamshidi et al. (2013) stated research in cloud migration, is in the early stages of maturity and further research is required in this area. A systematic literature search deals with identifying, categorizing and synthesizing the previous published studies and enabling knowledge sharing in the research community. This research provides a comprehensive framework for migration to the cloud by the meta-synthesis approach and systematic literature search of the published studies.

The remainder of this paper is structured as follows: Section 2 contains a description of the background. A general review of related research works and some critics of the previous published studies are given in this section. Section 3 describes research method that is adopted to conduct the current research. In this section, a systematic literature search, selecting appropriate research articles, synthesize qualitative findings and validation of research are explained. Section 4 describes our proposal. At first, we present preliminary framework, then we improve it and propose a comprehensive framework for cloud computing migration and cloud migration maturity model (CIM3). In Section 5, we describe research limitations and threats to validity. Finally, Section 6 concludes this paper.

2. Background and Related Work

This section discusses prior research in cloud computing migration. Khan and Al-Yasiri (2016) show that the lack of knowledge about cloud computing is one of the main obstacles in adoption and migration to the cloud. Alabadi (2011) investigates one of the main decision makers' concerns which is how to prioritize and select the appropriate cloud services to migrate toward cloud environment. He classifies activities of an organization from the perspective of sensitivity of the mission. Accordingly, Kundra (2011) offers "value" and "readiness" table to select applications and services for transmission. Based on Turban decision-making process, Alkhalil, et al. (2014) considered the design phase and business intelligence components. They present the knowledge-based decision support system (KBDSS), helping organizations in seamless migration to the cloud. Alkhalil's models, not only pay attention to the cost like others but also create a knowledge-based decision-making system. Andrikopoulos, Strauch, and Leymann (2013), Andrikopoulos, Song, and Leymann (2013) pose decision support systems too. In a similar way, Khajeh-Hosseini et al. (2011), Greenwood et al. (2010) also pose decision support tools.

In a different way, some authors such as Alonso et al. (2013), Menzel and Ranjan (2012), Johnson and Qu (2012), Pfizmann and Joukov (2011), Hajjat et al. (2011), Lewis et al. (2005), Jermyn et al. (2014) formulate

the cloud migration problem using an objective function including cost, revenue, net present value, rate of investment, etc. and they finally solve it to find its optimal solution.

Some of research focused on the selection of best cloud service provider, suitable solution and best cloud platform. García-Galán et al.(2016) present “automated analysis of feature models” (AAFM) to analyze and select the infrastructure. Garg et al. (2013) develop a framework based on Service measurement index (SMI). This framework uses the analytic hierarchy process (AHP) method to help customers to find the most appropriate cloud provider. Also, Omerovic et al. (2013), Chan and Chieu (2010), Li et al. (2010) have done more research on cloud service provider selection.

Phaphoom et al. (2015) and Surendro and Fardani (2012) have considered the factors that affect cloud computing adoption and migration. Rockmann et al. (2014) define three dimensions of required IT capabilities including technological, human, organizational capabilities to implement cloud computing technology in an organization.

Gholami et al. (2016) have investigated the cloud migration process and proposed an evaluation framework to classify approaches applicable to cloud migration. Hwang et al. (2016) present cloud migration orchestrator (CMO) which includes four engines includes discovery, analytics, configuration and migration. This framework automates and coordinates the cloud migration based on the IBM business process management (BPM) technology. With systematic literature review of migration to the cloud in 23 papers by Jamshidi et al. (2013), they propose a migration model including processes of migration planning, migration execution and migration evaluation. In their proposed model, cross-cutting concerns also cover main processes in an umbrella form. Andrikopolos et al. (2013) define four types of migration and introduce related adaption activities according to each type in different layers (i.e. data layer, business layer and the presentation layer). Guillén et al. (2013) for developing cloud applications propose a service-oriented framework that allowed application construct from combination of software components and migrate freely between cloud platforms. Menychtas et al. (2013) based on ARTIST⁴ methodology, present a new approach for modernization and adaptation of legacy application to cloud environment, with taking into account the technical and economical aspects.

As Gholami et al. (2016) state lack of a unified process model of the cloud migration is one of the main open challenges. Appendix A, Table A1 presents more information about reviewed papers. In Table 1, the migration steps from different published studies are shown.

Table 1
Migration steps from different published studies.

Authors	Migration steps
Khan and Al-Yasiri (2016)	Cloud requirement stage (CRS) - cloud preparation stage (CPS) - cloud migration stage (CMS)
Yaghmaei and Binesh (2015)	Developing the knowledge base about cloud - evaluating the present stage of IT (university) - experiment the cloud computing solutions - choosing the cloud computing solutions - implementation and management of the cloud computing solutions.
Rai et al. (2015)	Feasibility study- requirements analysis and migration planning-migration execution- testing and migration validation- monitoring and maintenance.
Wielki (2015)	Preliminary assessment- migration's plan creation- implementation and maintenance.
Pardeshi (2014)	Preparation- analysis- migration to cloud platform- concluding the cloud migration- maintenance and vendor management.
Kiadehi (2014)	Decision making- preparation- establishment- delivery and monitoring.
Abderrahim and Choukair (2014)	Baseline architecture- target architecture- migration decision- migration strategy- integration concerns.
Okai et al. (2014)	Planning- choosing the right deployment model- choosing the most suitable service delivery model- vendor selection- negotiating the SLA(service level agreement) - migration
Jamshidi, et al. (2013)	Migration planning- migration execution- migration evaluation.
Chauhan and Babar (2012)	Identification of requirements- identification of potential cloud hosting environments- analyzing application compatibility with potential cloud environments-identification of potential architecture solutions- evaluation of cloud platforms- evaluation of potential architecture solutions-implementation and system refactoring.

⁴ Advanced software-based service provisioning and migration of legacy software

With the holistic view, cloud computing migration consists of three stages called pre-migration, migration, and post-migration. Reviewing organizational failures in the implementation of applications and IT-based technologies reflects the fact that most life cycle models suffer from lack of a transparent stage of post-migration, while this stage is the longest phase of life cycle (Jaspersen, Carter, & Zmud, 2005). Therefore, complete review of the migration and post-migration processes are vital in cloud-related models because the success of its implementation is based on the acceptance, routinization and infusion (i.e., maximum usage of potential) of the technology (Saga and Zmud, 1993).

By reviewing the resources in the field of cloud computing, it is clear that a great deal of knowledge has been created related to the adoption of informational systems and cloud computing technology by many researchers, such as Davis (1989), Venkatesh et al. (2000;2003;2012), Tornatzky et al. (1990), El-Gazzar et al. (2016), Sabi et al. (2016), Gangwar et al. (2015), Safari et al. (2015), Prasad et al. (2014), Premkumar (2003). But a few researchers comprehensively have studied behaviors of post-adoption (migration) and post-migration. According to the selected studies in this research, some critics of the previous research are as follows:

❖ **A few usage of theoretical foundation**

As Gholami et al. (2016) in their survey research state, in more than 65% identified papers (28 papers from 43 papers), theoretical foundation is not applied. In retained papers, model-driven development, optimization techniques, usable techniques in software development (such as software product line, software pattern), etc. are mentioned as theoretical foundation. With reviewing the selected papers, we understand unlike the previous studies in the field of cloud computing adoption, a small number of published studies supported their research by a theory in the field of migration and implementation of cloud. Technical view is rather dominant on research; along with shortage of theoretical framework.

❖ **A few comprehensive qualitative research**

According to the selected studies in this research, the most studies emphasize quantitative approaches while the shortage of comprehensive qualitative research such as qualitative meta-synthesis method (focuses on the comprehensive review and synthesis of findings) is tangible. This fact is reported in other published studies such as Gholami et al. (2016) who state in a few researches, semi-structured interview is applied as a qualitative research method and only fewer than 14% identified papers, defined their research process.

❖ **Lack of holistic view**

Most of previous studies lack a systematic review and presentation of holistic view. Majority of Proposed approaches incorporated with heterogeneous technical-centric concept. By rely only on the earlier studies, managers don't find the comprehensive framework of cloud computing migration, in which all cases are defined.

❖ **Static consideration of migration and implementation processes**

Most studies consider the migration as static processes. They have not considered maturing stages of cloud computing in implementation and migration.

In the next Sections a new framework is proposed to overcome to the above mentioned problems.

3. Research method

The model of the research process that was provided by Oates (2006) is a suitable reference for the present research method. According to the aforementioned model, firstly, researchers must define the component that make up the research process including strategy (e.g., survey, design and creation, experiment, case study, action research, ethnography), data generation methods (e.g., interview, observation, questionnaires, documents) and data analysis methods (quantitative or qualitative). In the present research, "Design and Creation" is taken into account as the research strategy. The data generation method is "documents" and records of the previous resources. In this study, qualitative meta-synthesis is used to analyze data and combine results from a variety of published studies.

"Design and creation" strategy consists of five stages which are called *awareness*, *suggestion*, *development*, *evaluation* and *conclusion*(Oates, 2006). In the *awareness* stage, the problem of cloud computing migration is described by systematic literature search, also new areas and dimensions are defined. In *suggestion* and *development* stages, meta-synthesis approach is applied to identify concepts, related sub-categories and related categories and ultimately provide a comprehensive framework. In the *evaluation* stage, validation of the

proposed framework is performed by the opinion of experts (seven experts have been purposefully selected by using the snowball technique in the form of a focus group). A *conclusion* also constitutes the final stage. In this stage the acquired knowledge along with design process explanation is documented.

In fact, to provide a framework for cloud computing migration, it is needed to determine main components by reviewing the previous published studies. For this reason, we use the meta-synthesis method that is one of the qualitative methods and it is consistent with interpretive paradigm assumptions. The meta-synthesis method has not been used in cloud computing migration studies yet. By providing a systematic approach for researchers, meta-synthesis method explores new and essential concepts through synthesizing qualitative research. Meta-synthesis method creates a comprehensive and wide spread view toward the problems in addition to promote the current knowledge. It also is a method of reinterpreting and reshaping existing qualitative findings (McClean and Shaw, 2005). A qualitative meta-synthesis is a technique that combines results from a variety of studies with a common theme. Per se, “The sample for a meta-synthesis, then, is made up of individual qualitative studies selected on the basis of their relevance to a specific research question posed by the synthesist” (Zimmer, 2006).

In carrying out a qualitative meta-synthesis method, there are three specific objectives: theory building, theory explication and theory development (Douglas et al., 2008). The objective of this research is to explicate the theory, which a more holistic view of the phenomenal results from the synthesis of qualitative findings from previous studies.

There are various meta-synthesis Methods. Perhaps the first planned synthesis of qualitative findings is done by Glaser and Strauss (Zimmer, 2006). Noblit and Hare offer a seven-step method based on qualitative ethnography (Zimmer, 2006). Sandelowski and Barroso (2007) also propose a seven-step method that has more generality and it is the selected method of this research. These steps include: formulate the review questions, conduct a systematic literature search, screen and select appropriate research articles, extract the results, analyze and synthesize qualitative findings, maintain quality control and present findings. According to all aforementioned explanations, the process used in this research is summarized in Fig. 2 (these steps will be explained later in different sub-sections).

As shown in Fig.2, we have defined a feedback process. As it will be explained later in Section 3.4, we used experts' peer review to validate the proposed framework. In addition, to optimize the validity of qualitative research process, we used two independent reviewers who should answer to predefined questions in different steps of our research process. As a result, the comments received from the focus group members (i.e., the experts) and two independent reviewers can affect on each step of the research process and modify it. Therefore, if the experts and reviewers recognize that one of the steps of the research process has not been done in a right way; this step must be done again.

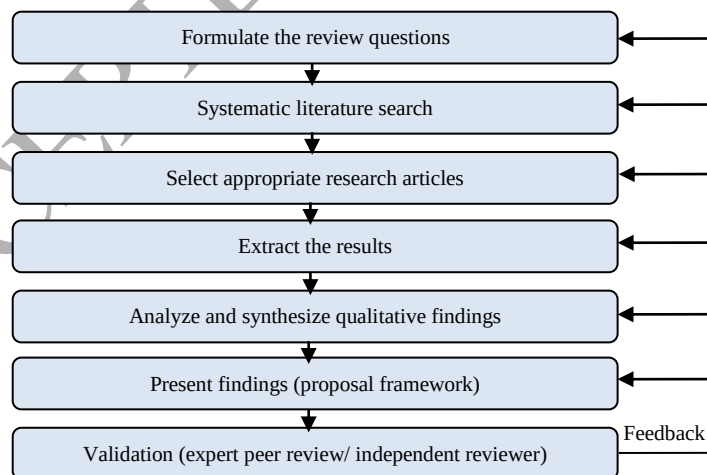


Fig.2. The research process.

3.1. Formulate the Review Questions (RQs)

With the holistic view and systematic search of previous published studies, this research looks to eliminate the mentioned deficiencies (as explained in previous section), and find categories and concepts related to cloud

computing migration in the previous published studies. The research answers the following questions and provides the comprehensive and dynamic framework of cloud computing migration using meta-synthesis method.

- RQ1. What are the existing processes or phases of cloud migration?
- RQ2. How do these processes or phases form migration framework?
- RQ3. Based on our research, what can we conclude about the appropriate cloud migration framework?

3.2. Systematic Literature search and Selecting Appropriate Research Articles

All articles from relevant journals and conferences from 2000 (the obtained results from our primary search indicated that the published studies before the year 2000 can be negligible) to the first half of 2016 in different databases and search engines such as Elsevier, Scopus, Springer, IEEE Xplore and Google Scholar are statistical population of this study. Keywords of cloud computing, migration, migrate, acceptance, usage, and implementation are applied for searching. The Criteria considering articles as part of the research are shown in Table 2.

Table 2
Criteria for considering as part of the research.

Criteria No.	Parameters	inclusion criteria	exclusion criteria
C1	Paper's language	Studies written in English	Studies not written in English
C2	Time of presenting papers	Papers published from 2000 to the first half of 2016	Studies published prior to 2000
C3	Research subject	Migration to cloud computing environment	Cases different from the mentioned subject
C4	Type of study	Papers published in peer reviewed journal and international relevant conferences	Book reviews, non-peer reviewed journals, personal comments
C5	Status of research information	Papers with clear research process and research findings	Papers with unclear research process and research findings

In early search, based on keywords we found 657 related articles. After investigating the title it is determined that most of them are related to other research fields of cloud computing and 145 articles are selected for reviewing abstract. After studying the abstracts, 63 articles are chosen for full content review. After a full review of content, 11 articles are rejected and finally 52 articles are obtained for analysis of content (see Appendix A, Table A1). The number of selected papers with separate years of publishing, relevant databases and search engines and also the process of searching and selecting appropriate articles are shown in Fig. 3, Fig.4 and Fig. 5 respectively.

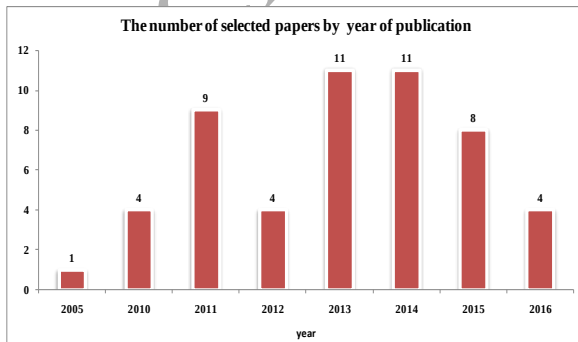


Fig.3. The number of selected papers by year of publishing.

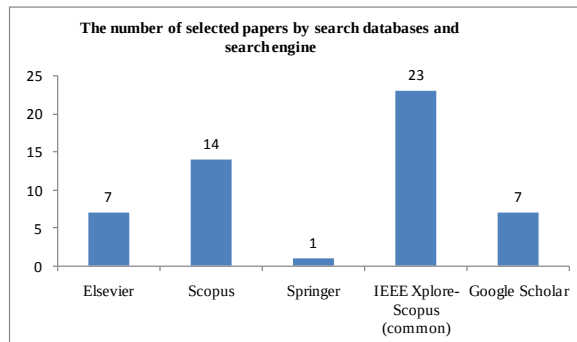


Fig.4. The number of selected papers by search databases and engines.

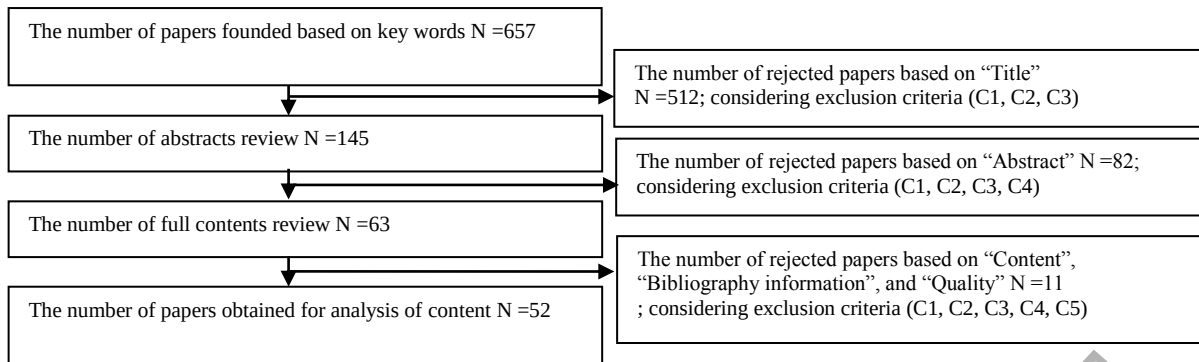


Fig. 5. The process of searching and selecting appropriate articles.

3.3. Extract The Results, Analyze and Synthesize Qualitative Findings

For extracting the concepts from the text of the research questions, two following questions have been simultaneously taken into account.

- RQ1. What are the existing processes or phases of cloud migration?
- RQ2. How do these processes or phases form migration framework?

In many papers in the literature review, the concepts have been directly mentioned. We extract these concepts from the literature review and write them with their references in Appendix B, Table B1. Then, we gather the similar concepts together based on the meaning of each concept and after that we extract the related sub-categories. This procedure is repeated in creating categories which have been derived from similar sub-categories. Concepts, sub-categories and categories are validated by the experts' review. Finally, the main phases of the initial framework are determined based on created categories. By comprehensively studying the resources, 91 concepts are extracted and accordingly, 13 sub-categories and 6 main categories are grouped (see Appendix B, Table B1).

As an example of extracted concepts, sub-categories and categories; we show a category titled "Initiation phase", its sub-categories and related concepts in table 3. This category is composed of twelve various extracted concepts and two related sub-categories. Concepts are taken out from various papers and considering to their similarity in meaning we classify them in two sub-categories.

Table 3

Example of concepts and sub-categories as well as their extraction resources related to initiation phase.

No	Concepts	References	Sub-categories	Category	
1	Developing and expanding the knowledge about cloud computing.	(Khan and Al-Yasiri, 2016), (Yaghmaei and Binesh, 2015), (Pardeshi, 2014), (Alkhalil et al., 2014)	Expanding the knowledge about cloud computing and their basic requirements	Initiation phase	
2	Changing the mentality.	(Kiadehi, 2014), (Rewatkar and Lanjewar, 2010), (Kundra, 2011)			
3	Understanding the legal requirement and implication related to the cloud.	(Alsufyani et al., 2015)			
4	Should become fully aware of the cloud computing principles.	(Alkhalil et al., 2014)			
5	Reviewing the standards related to the cloud.	(Khan and Al-Yasiri, 2016), (Kiadehi, 2014)			
6	Become fully aware of the cloud contract terms.	(Khan and Al-Yasiri, 2016)			
7	Identifying the services and business processes.	(Yaghmaei and Binesh, 2015), (Rockmann et al., 2014)	Identification of opportunities and threats		
8	Identifying the cloud environment.	(Alkhalil et al., 2014)			
9	Identifying opportunities.	(Alkhalil et al., 2014)			
10	Identifying limitations (constraints).	(Khan and Al-Yasiri, 2016), (Alkhalil et al., 2014), (Alsufyani et al., 2015)			
11	Identifying risks.	(Khan and Al-Yasiri, 2016), (Omerovic et al., 2013), (Alkhalil et al., 2014), (Kiadehi, 2014), (Khajeh-Hosseini et al., 2010b)			
12	Developing knowledge base.	(Andrikopoulos et al., 2013b), (Alkhalil et al.,			

	2014)		
--	-------	--	--

3.4. Validation of Research

The consensus of most researchers is that the output of meta-synthesis method can be validated by one of the following methods:

- Experts peer review in the approval of research outputs (Sandelowski and Barroso, 2007)
- Providing a comprehensive result of using new case studies (Thorne, 2009)

In this research, Experts' peer reviews in the acknowledgement of research achievements are used to validate the proposed framework. Therefore, seven experts (independent people of the research team who do related research or have an activity in the industry and related fields) associated with cloud computing in Iran, are selected by purposive and snowball technique, then we have arranged some focus group meetings. Finally the proposed framework is finalized and confirmed.

The experts of this research have evaluated and validated our proposed cloud computing migration framework. It is very important to state that the selected experts in focus group must be knowledgeable in the field of cloud computing technology. Accordingly, we categorized the experts' experiences in two main groups. The members of first group have academic activities related to cloud computing technology and have published relevant research papers. The second group, work in cloud computing industry as cloud service providers. The first group's members focus on the proposed framework from academic point of view while the second group mainly considers the technical aspects. These groups have a close collaboration in our focus group.

A key mechanism for optimizing the validity of study is audit trail - consist of documents tracking search- (Sandelowski & Barroso, 2007). We maintain of an audit trail. The research documents are delivered to reviewers and focus group members. It is worthy to note the focus group consists of research team and seven experts.

For optimizing the validity of qualitative research synthesis studies, according to Sandelowski and Barroso (2007) procedures; research team meet together weekly (for 5 months and 2 meeting with 8 hours per week) to discuss and formulate outcomes and refine study appraisal strategies. All procedures are documented (audit trail) and delivered to two independent reviewers (different from the focus group), who have great rich experiences in qualitative research, to appraise each report and evaluate research process. In different situation, they should answer to predefined questions:

- Are meta-synthesis research processes done in a right way (by research team)?
- Paper selected for research, are related and suitable? If papers selection process is done in a right manner or not?
- Whether the selected papers cover the research topic or not?
- Are the independent reviewers' opinions about rejected papers close to research team's opinion? (independent reviewers randomly appraise some rejected papers and response to this question)
- If the independent reviewers' opinion about extracted concepts, sub categories and categories are close to research team's opinion or not? (independent reviewers randomly appraise some papers and extract concepts then compare their result to research team result)

4. Results (Proposed framework) and Discussion

In this section, we first propose a primary framework. This primary framework is constructed directly based on the extracted concepts, sub-categories and categories (i.e., the result of qualitative Meta-synthesize research methodology as shown in Appendix B, Table B1). It should be noted that to avoid probable misguiding, it seems better that the readers should first consider this primary framework. Then, we improve the primary framework based on complementary findings and add new phases (categories) to it as well as proposing the final migration framework. Finally, we clarify our proposed cloud migration maturity model for improving migration processes.

4.1. Preliminary framework

According to the concepts, sub-categories and categories of Table3 which are the results of qualitative analysis and synthesis of research source's findings, the following preliminary framework (Fig. 6) is recommended for stepwise migration to cloud computing environment. The mentioned framework consists of six main phases (categories) including *initiation*, *adoption*, *decision-making and selection*, *migration*, *adaptation and control*, *routinization and maintenance* stages and thirteen sub-phases.

Initiation phase:

In the initial phase as Cooper and Zmud (1990) explained, organizational need or technological innovation or both of them increase the pressure for change. Organizations deal with proactive investigation of problems, opportunities and adoption of cloud solutions related to them. One of the main obstacles in the way of organizations' migration to cloud computing environment is the lack of awareness of top managers and decision-makers from cloud computing. For a successful start, organizations must effectively extend their cloud knowledge (Khan and Al-Yasiri, 2016; Yaghmaei and Binesh, 2015; Pardeshi, 2014; Alkhalil, et al., 2014). In addition, it is necessary for organizations to modify their mentality from infrastructure ownership toward the providing services. Organizations should understand legal considerations, verify standards and develop knowledge database (Alkhalil et al., 2014; Alsufyani et al., 2015; Kiadehi, 2014). They should identify the cloud environment, opportunities, risks and threats (Alkhalil et al., 2014; Khajeh-Hosseini et al., 2010a; Omerovic et al., 2013; Rockmann et al., 2014). The more accuracy in the process of acquiring knowledge leads to more effective analysis process at later phases.

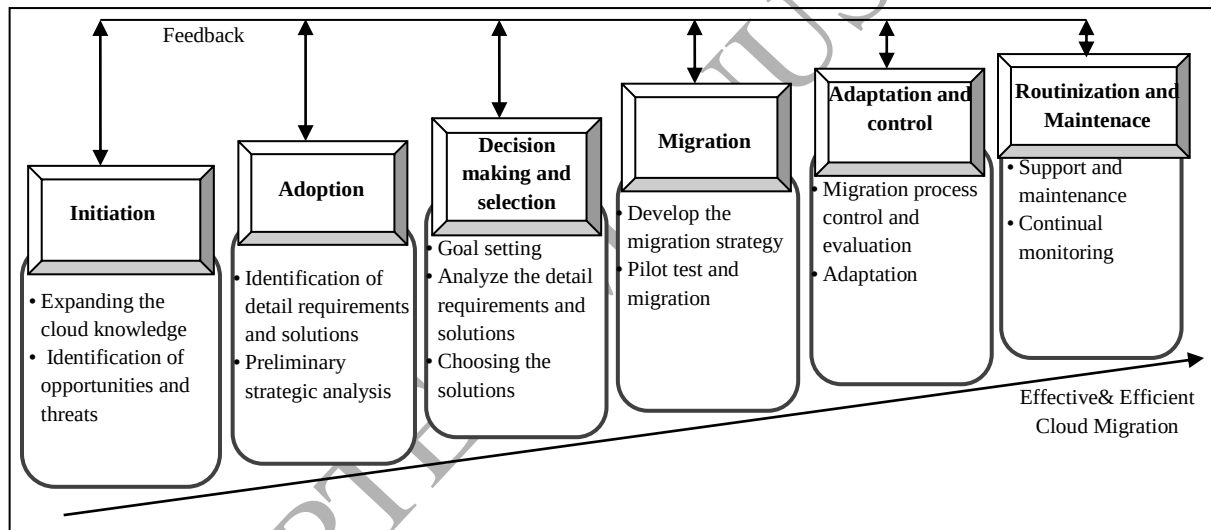


Fig. 6. Preliminary stepwise migration framework to cloud computing environment.

Adoption phase:

In the adoption phase, according to the opinion of Cooper and Zmud (1990) and Kwon and Zmud (1987) a logical and political bargaining in the organization leads to organizational support to implement IT applications. At this phase, the organization incline to cloud computing migration, and the decision is made to invest in the required resources. For reasonable bargaining in the organizations, separate needs and requirements should be firstly identified fully-detailed. In addition to fully understanding of applications (Menychtas et al., 2013), the requirements of stakeholders, compatibility, security (Subramanian and Seshasaayee, 2014), the potential cloud hosting environments and the proposed solutions are needed to be accurately and completely identified (Chauhan and Babar, 2012). After the identification step, preliminary strategic analysis is required to be conducted. In this step the value of cloud computing is analyzed (Kundra, 2011). After enterprise readiness identification (Khan and Al-Yasiri, 2016; Kiadehi, 2014), organizational strengths and weaknesses, opportunities and threats, in the previous step, the market landscape (Kiadehi, 2014), business strategy (Alkhalil et al., 2014), technology vision, IT governance status (Khan and Al-Yasiri, 2016) and specification of governmental service should be evaluated.

Decision-making and selection phase:

At this phase, after the preliminary strategic analysis and evaluating the positivity of organization's opinion toward cloud computing migration, migration goals must be set. Then, deployment model, service model, bench points, and architecture of target should be determined (Menychtas et al., 2013; Omerovic et al., 2013; Pardeshi, 2014). It is notable that being align with the organizational goals and strategies should always be considered in setting the goals by IT managers. After goal setting, stakeholder requirements, required IT capabilities (technical, human and organizational) and outsourcing capabilities (ITO) should be analyzed (Rockmann et al., 2014). Technical and economical feasibility analysis is should be done (Alonso et al., 2013; Jamshidi et al., 2013; Johnson and Qu, 2012; Khan and Al-Yasiri, 2016; Pfitzmann and Joukov, 2011), also suitability of cloud-based services should be evaluated with the potential cloud environment (Alkhalil et al., 2014; Chauhan and Babar, 2012; Garg et al., 2013). By developing the knowledge-based decision support system from the analysis, which is done in the previous step, organizations can rank services and service providers based on various criteria, including the user's previous experience and performance of the service. Organizations can choose a suitable provider for migration by selecting the appropriate platform (Garg et al., 2013; Menzel and Ranjan, 2012).

Migration phase:

The most important action on the migration phase is developing the migration strategy. Then, to apply the developed strategy, it is better to choose a pilot project and proceed to migrate after the initial test. In this phase, it is needed to make policy and plan for cloud migration, and doing the categorization and assignment of responsibilities according to that.

Services, applications and data should be classified along with ones which should migrate must be selected according to various criteria (such as: high updating, high maintenance costs, low utilization rates, high cost per user and innovation of delivered service). Then, decision makers should classify selected application and services based on sensitivity of the mission (Alabbadi, 2011). We suggest applying traditional system for highly sensitive application with critical missions (not to be migrated). If organizations want to use cloud solutions for mentioned services and application, it is better for them to use private cloud deployment model. Public cloud is Suitable for low sensitive applications and services.

For effective migration, interoperability strategy (Abderrahim & Choukair, 2014), multi-tenancy and elasticity strategy (Andrikopoulos et al., 2013c), outsourcing strategy (Pardeshi, 2014), exit strategy and transmission schedule (Kiadehi, 2014) should be developed along with determination of immigrant services. After migration strategy development, modernization and adaptation of the legacy applications are done (Menychtas et al., 2013) and effective contract (Khan and Al-Yasiri, 2016) is presented. To perform migration process, it is better to select a pilot project at first, and employ systems' testing, retrieval and adaptation of the architecture (Jamshidi et al., 2013) and performing extraction and migration of data, services and applications as a seamless and integrated migration (Alkhalil et al., 2014).

Adaptation and Control phase:

The purposes of this phase are control, monitor and evaluate important issues during migration, and then adapting the various factors to reach the desired results. By controlling the application for its security, availability and performance throughout the migration process, this phase looks to ensure continuity of the application and integration of the new system with the older system (Abderrahim and Choukair, 2014; Jamshidi et al., 2013; Menychtas et al., 2013). In order to realization of technical and economical goals which are set in the adoption phase, it should be evaluated and verified during and after the migration. For improvement, adaptation activities should also be done during and after the migration, such as the acquisition of new skills depending on the needs (Kundra, 2011), review of the existing IT governance model and creation of a governance model related to organizational strategies; also all relevant policies are reviewed for ensuring effective support (Kiadehi, 2014). As Cooper and Zmud (1990) also express, in adaptation phase, organizational procedures are reviewed and developed, besides organization's members are trained toward new procedures and IT applications (Cooper and Zmud, 1990).

Routinization and Maintenance phase:

At this phase, in addition to the creation of guidelines, activities related to support, update and vendor management are conducted. The routine budget is considered for migration. Cooper and Zmud (1990) and Kwon and Zmud (1987) define routinization phase in such a way that usage of systems and IT application

are part of individual's routine behavior, normal activities and governance systems of organizations have been adjusted to respond to IT application. At this phase, in order to respond to organization's governance systems, evaluation of service delivery and vendor's models (Kundra, 2011) are periodically and continuously done as well as routine monitoring of quality of service (QoS) (Kundra, 2011; Pardeshi, 2014) and the way of meeting service level agreement (SLA) (Garg et al., 2014).

4.2. Improving the preliminary framework and propose comprehensive cloud migration framework

Apart from knowing how the innovative technology are acquired by the organization (created internally or purchased commercially), they should be incorporated since their benefits are fully flourished (Hazen et al., 2012). In fact, there is a gap between the adoption of innovative technology and its incorporation in the organization so that it must be filled with *acceptance*, *routinization* and *infusion* (or, alternatively, *assimilation*) stages (Hazen et al., 2012).

In order to better understanding of *adoption*, *acceptance*, *routinization* and *infusion* (or, alternatively, *assimilation*) stages, it is better to refer to the six-step model of IT implementation presented by Cooper and Zmud (1990) and Kwon and Zmud (1987). They have introduced a six-stage model of IT implementation includes *Initiation* (organizational needs and innovation will lead to increase in pressure to change), *Adoption* (a logical and political bargaining in the organization will lead to decide for investing in IT application adoption), *Adaption* (IT application is developed, installed and maintained; also organizational procedures are reviewed and developed), *Acceptance* (reflects users' commitment for the use of new system), *routinization* (using system and IT application are parts of individual's routine behavior and normal activity), *infusion* (it mentions incorporation and the maximum usage of application's potential in a deep, broad and comprehensive way in the organizational or individual work system).

So, to effectively use of cloud computing migration framework, it is better to entirely consider IT implementation model. The primary framework covers the first five stages of IT implementation model include *initiation*, *adoption*, *adaptation*, *acceptance* and *routinization* but there are not any phases about *infusion* stage (as shown in Fig. 6). Indeed, *infusion* stage means the incorporation and the maximum usage of technology's potential in a deep, broad and comprehensive way in the organizational or individual work system. Accordingly, to improve the preliminary framework, it is necessary to add a new phase (including *infusion* stage). Thus, it is suggested to add "*optimizing and Infusion*" phase to the previous framework (as shown in Fig.7) and define a new model for the maturity of cloud computing migration processes.

The *optimization and infusion* phase points to the dynamics of migration process and its perfection trend. In the *optimization and infusion* phase, the migration processes are optimized in line with organizational strategic objectives. As Cooper and Zmud (1990) stated, the infusion phase points out deep, widespread and comprehensive implementation and incorporation of any technology such cloud computing technology in the organization. Also, technology is used to its utmost potential in the organization. Then, with applying this new added phase, organizations ensure that the usage of cloud computing technology reaches to the maximum possible value in terms of related volume and diversity of organizational processes, and depth of usage.

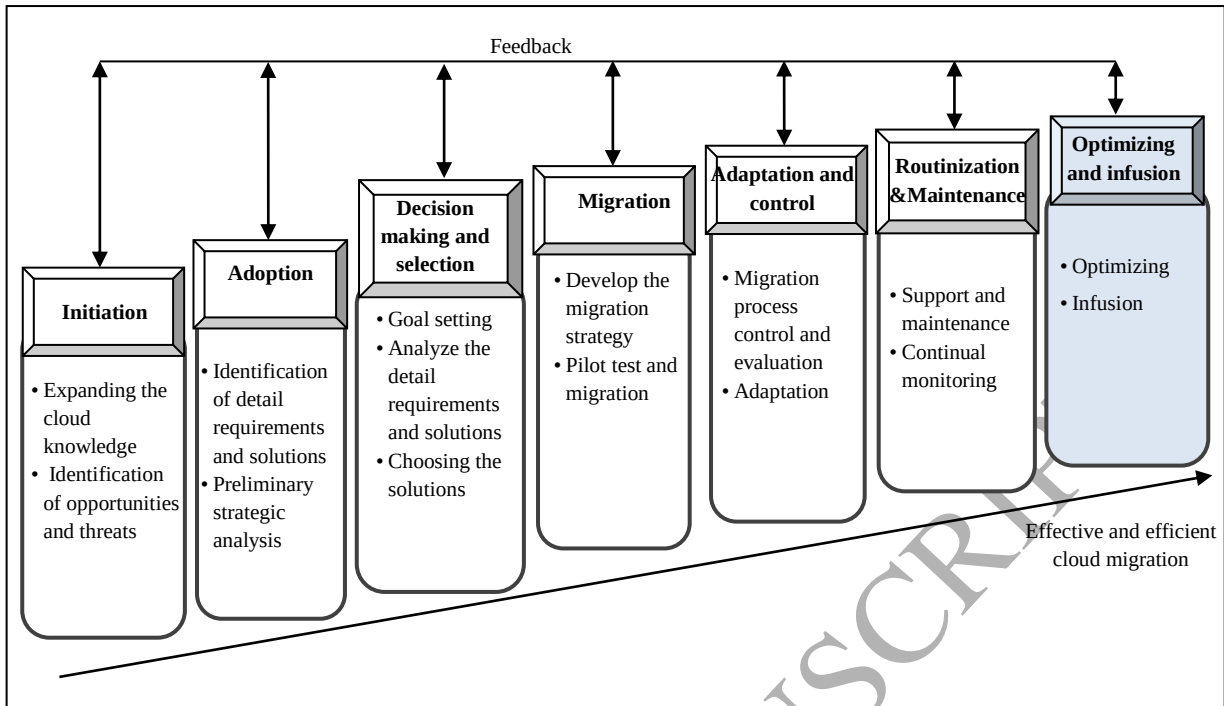


Fig. 7. Comprehensive cloud migration framework.

4.3. Cloud Migration Maturity Model (CIM3)

If the proposed framework is considered with process approach, its phases consist of migration processes and sub-processes which the migration maturity model can be defined for improve them. Using maturity model guide the organization to a level of perfection in the migration process by going through different levels of it. To define the cloud migration maturity model, we review different maturity model such as CMMI, OPM3, BPMM, etc. Finally, inspired by representation approaches (staged representation and continuous representation) introduced in CMMI for Services ver. 1.3 (Team, 2011); we construct our proposed “Cloud Migration Maturity Model” (CIM3). In this maturity model, organizational processes scope only contains cloud migration processes (as shown in proposed migration framework (Fig. 7)). Our main reason for selecting CMMI is its application in organizations from various industries (Zubrow, 2003).

Cloud migration maturity model (CIM3) helps organizations find out the right chose for migration to cloud computing environment by building capability in their cloud migration processes. To validate this maturity model, we used new focus group meetings with previous members who review, evaluate and finalize it for us. CIM3 contains the following model components:

- Migration process areas: CIM3 is a collection of defined seven migration process areas and fifteen related processes that presented in proposed framework (Table 4). According to CIM3, a migration process area is satisfied when it covers all of the generic and specific goals and practices for that process area.
- Generic goals and generic practices: they are general categories of goals that may fulfill certain needs. They are part of every migration process area (Appendix C, Table C1).
- Specific goals and specific practices: they are specific to a migration process area (Appendix D, Table D1).

The CIM3 is structured as two representation approaches in which just one of them must be selected. Note that, both CIM3 representations contain above mentioned model components.

- Continuous representation (capability levels): this approach allows an organization to choose one or more specific cloud migration process areas and improve relative to these. It uses capability levels (level 0 to level 3) to distinguish improvement relative to an individual process area. Attribute of each capability level, required generic goals and generic practices is shown in Table5.
- Staged representation (maturity levels): it is an approach that uses sets of migration process areas to define an improvement path for an organizational cloud migration. This improvement path is described

by five maturity level. Description of each maturity level in CIM3 staged representation is shown in Table 6.

In fact, if organizations want to use CIM3, they should select one of the two representation approaches in the first step. A representation approach allows an organization to follow different improvement objectives. Note that in continuous representation approach, organizations can select one or more migration process and improve their capabilities by covering all generic goals and their related specific goals (see Table5). But, Staged representation approach uses sets of migration process areas to define an improvement path for an organizational cloud migration. This improvement path is described by five maturity level with their required generic and specific goals that should be covered (see Table 6 and Table 5).

Table 4
Migration processes in CIM3.

No	Abb.	Core processes	Process area
1	ECK	Expanding the cloud knowledge	Initiation
2	IOT	Identification of opportunities and threats	Initiation
3	IDR	Identification of detail requirements and solutions	Adoption
4	PSA	Preliminary strategic analysis	Adoption
5	GSE	Goal setting	Decision making and selection
6	ADR	Analyze the detail requirements and solutions	Decision making and selection
7	CSO	Choosing the solutions	Decision making and selection
8	DMS	Develop the migration strategy	Migration
9	PTM	Pilot test and migration	Migration
10	MPC	Migration process control and evaluation	Adaptation and control
11	ADA	Adaptation	Adaptation and control
12	SAM	Support and maintenance	Routinization and maintenance
13	CMO	Continual monitoring	Routinization and maintenance
14	OPT	Optimizing	Optimizing and infusion
15	INF	Infusion	Optimizing and infusion

Table 5
Capability levels within CLM3 contiguous representation.

Capability levels	Attributes	Status	Generic goals	Generic practices
CL0 (Incomplete)	- Ad hoc process	- Failure to reach CL1	Process is either not performed or only partially performed in that one or more of the specific goals for a process area at level 1 are not satisfied.	
CL1 (Performed)	- A process that accomplishes the work necessary to satisfy the specific goals.	- All specific goals of the process area are satisfied	GG1: Achieve Specific Goals	GP 1.1: Perform Specific Practices.
CL2 (Managed)	- A performed process that is planned and executed in accordance with policy - Employs skilled people - Is monitored - Involves Stakeholders	- All specific goals of the process area are satisfied - Institutionalize a Managed Process	GG2: Institutionalize a Managed Process	GP 2.1: Establish an organizational policy GP 2.2: Plan the process GP 2.3: Provide resources GP 2.4: Assign responsibility GP 2.5: Train people GP 2.6: Manage configurations GP 2.7: Identify and involve relevant stakeholders GP 2.8: Monitor and control the process GP 2.9: Objectively evaluate adherence GP 2.10: Review status with higher level management
CL3 (Defined)	- A managed process that is tailored from the organization's set of standard processes. - has a maintained process description - contributes process related experiences to	- All specific goals of the process area are satisfied - Institutionalize a managed process - Institutionalize a defined process	GG3: Institutionalize a defined process	GP 3.1: Establish a defined process GP 3.2: Collect improvement information

Capability levels	Attributes	Status	Generic goals	Generic practices
	organizational Process assets.			

Note that in continuous representation, organization can select one or more migration process and improve their capabilities by covering all generic goals and their related specific goals.

Table 6

Description of each maturity level in CLM3 staged representation.

Maturity levels	Description	Core processes	Processes capability level
ML1	- Migration processes are unpredictable, poorly controlled and reactive. - The migration process performance may not be stable and may not meet specific objectives such as quality, cost, and schedule, but useful work can be done.	-	-
ML2	- Migration processes are planned, documented, performed, monitored, and controlled at the project level (often reactive). - The managed migration process comes closer to achieving the specific objectives such as quality, cost, and schedule.	All migration processes (initiation phase to adaptation phase) are managed process.	CL2 (managed processes) *
ML3	Migration processes, standards, procedures, tools, etc. are defined at the organizational level. Project or local tailoring is allowed, however it must be based on the organization's set of standard processes and defined per the organization's tailoring guidelines.	All processes (initiation phase to adaptation phase) are defined processes.	CL3 (defined processes)*
ML4	Migration process is measured and controlled	All processes (initiation phase to adaptation phase) are defined processes. Routinization and maintenance processes are defined processes.	CL3 (defined processes)
ML5	Focus is on continuous quantitative improvement	All processes (initiation phase to Routinization phase) are defined process. Optimization and Infusion processes are defined processes.	CL3 (defined processes)

* Managed processes and defined process are defined in Table5.

Comparison between two defined representations in CLM3 is shown in Table7.

Table 7

Comparison of maturity levels and capability levels in CLM3.

Level	Staged representation (maturity levels titles)	Continuous representation (capability levels titles)
Level 0	-	Incomplete
Level 1	Initial	Performed
Level 2	Managed	Managed
Level 3	Defined	Defined
Level 4	Quantitatively Managed	-
Level 5	Optimizing	-

5. Research limitations

There are some limitations in this research (threat to validity) which are explained to clarify research process:

- Publication bias

A threat to the validity of integration studies like this research is including only published papers (Sandelowski & Barroso, 2002) and studies are written in English. This may lead to publication bias (Johansson, Fenwick, & Premberg, 2015)

- Bias in selecting appropriate research articles

This research focused on studies that reviewed the cloud computing migration. For controlling the mentioned bias and reducing missing relevant papers, we extend the research keywords (defined in Section 3.2). Furthermore, we search in primary selected papers references to find related papers which not necessarily use our defined keywords. Additionally, such as description in section 3.4, we consult

two independent reviewers to evaluate paper selection process. Finally, based on research methodology, 52 research papers selected (as shown in Appendix A, Table A1).

- Researcher bias

Every researcher has some kind of bias. The bias could be simply promoting one theory over other or neglecting to interview some kinds of disagreements. To control this bias and reduce threat to validity, research team use weekly discussion meeting. In this meeting, researchers freely discussed about defined topics. Additionally, we consult two independent reviewers to realize research team activities such as research method, paper selection, concepts extraction, result presentation, etc. Furthermore, extracted data with their references (as shown in Appendix B, Table B1) delivered to seven experts, and in a focus group meeting they are evaluated and finalized.

6. Conclusions and future work

New technologies such as cloud computing play an important role in organizations flexibility and productivity and allow them to take proper steps in developing and creating competitive advantage by accessing the capabilities of additional value. In order to take advantages of cloud computing infrastructure and services, organizations need to migrate to the cloud environment. Migration to the cloud is a strategic organizational decision with complex and dynamic nature. The benefit of this research is that, for the first time, has used the qualitative meta-synthesis method to analyze and synthesize the result of previous studies whose main objective was to recommend a framework or model for cloud migration. This research has founded the concepts, sub-categories and categories affecting cloud computing migration and provided a comprehensive cloud computing migration framework containing initial stages of pre-migration, migration and post-migration.

As RQ1 is concerned, research team uses systematic literature search on cloud migration related papers. According to research process (as shown in section 3) we search in related database and select 52 relevant papers to analyze and identify the cloud migration processes and phases. In regarding RQ2, to realize how these processes or phases form migration framework, we deeply review the selected papers and extract relevant concepts. To ensure doing right manner to extract appropriate concepts, we request two independent reviewers who are expert in qualitative research, comment on these processes. After several discussion meetings among research team, concepts validated and finalized.

In regarding to RQ3, we synthesize previous research findings and pose a suitable migration framework based on the finalized extracted concepts, sub categories and categories. For validating defined categories and pose preliminary framework, we use expert peer review method. Considering the all experts comments and finalizing the sub categories and categories (as shown in Appendix B, Table B1), preliminary framework is proposed. This preliminary migration framework has been constructed directly based on synthesizes of reviewed papers findings. If reviewers pursue the defined research process, without difficulty they validate this framework. Inspired by the IT implementation model introduced in (Cooper & Zmud,1990; Kwon & Zmud,1987) and according to complementary study done by research team, the preliminary framework improved and final cloud migration framework (Comprehensive cloud migration framework) presented (as shown in Fig. 7).

Additionally, we propose maturity model which is named “CIM3” for ensuring dynamism and improving in cloud migration processes. Our proposed framework and maturity model helps managers to gain a comprehensive overview of migration and they can perform strategic planning for its effective management. In connection with the future activities, it is suggested to use mixed method research and case studies to validate the proposed framework.

Acknowledgements

We would like to show our gratitude to the software development research group of *ICT research institute* and IT-department of the *Niroom Research Institute* (NRI) for sharing their pearls of wisdom with us during the course of this research. The authors also wish to thank “anonymous” reviewers for their so-called insights.

Appendix A

Table A1
Purpose and proposed output of selected research papers.

References	Purpose of research and proposed output
------------	---

References	Purpose of research and proposed output
(Khan and Al-Yasiri, 2016)	The proposed stepwise framework consists of cloud requirement, cloud preparation and cloud migration. They focus on preparation of organizations by informing them about risks. Studies show that the lack of expertise's knowledge about cloud computing is one of the main obstacles in the way of adoption and migration to the cloud.
(Hwang et al., 2016)	Presented Cloud Migration Orchestrator (CMO), includes four engines (parts) of discovery, analytics, configuration and migration. This framework automates and coordinates the cloud migration based on the IBM business process management (BPM) technology with pre-migration analytics.
(García-Galán et al., 2016)	The authors present automated analysis of feature models (AAFM) to analyze and select the infrastructure. Considering the infrastructure's requirements and costs, this model simplifies the configuration process and especially selecting model.
(Gholami, et al., 2016)	They survey the cloud migration process and propose an evaluation framework to classify approaches applicable to cloud migration.
(Kumar et al., 2015)	To reduce business data recovery time in disasters (man-made or natural) and minimize downtime for business process, they improve disaster recovery planning (DRP) based on the prioritization of data by using topsis technique, in order to storage data in IaaS.
(Tashkandi and Al-Jabri, 2015)	They Suggest improvement in network and internet infrastructure for providing services at reasonable and affordable costs. These are prerequisites for the acceptance and implementation of cloud computing and the privacy should be ensured.
(Alsufyani et al., 2015)	They propose, before migrating to the cloud several factors like, a deployment model (private or public), security (what type of data should go to the cloud) should be determined as well as customizable capabilities, legal considerations, and implications.
(Yaghmaei and Binesh, 2015)	According to their opinion, the implementation and usage stages of cloud computing in higher education institutions are formed with stages of Developing the Knowledge base about Cloud; evaluating the current stage of IT, experiment the cloud computing solutions, choosing the cloud computing solutions besides its implementation and management.
(Alsaffar et al., 2015)	For cloud screen-migration, they provide the internet protocol television (IPTV) service framework, based on secure authentication mechanism and lightweight content encryption method. The proposed framework prevents cyber-attacks and reduces vulnerability.
(Phaphoom et al., 2015)	The research is based on the logistic regression method on security and technical obstacles which prevent adoption of cloud computing services. Accordingly, data privacy, security and mobility are detected important.
(Rai et al., 2015)	In systematic literature review from 2009 to 2014, factors affecting on cloud migration and migration challenges are identified. Their study focuses on the necessity of creating a secure and comprehensive migration model to the cloud. The authors offer five-stage model for migration, includes feasibility analysis, requirements analysis and migration planning, migration execution, testing and migration validation, and monitoring and maintenance.
(Wielki, 2015)	He states that that 17.9% of young organizations and 10.2% of mature organizations (more than 5 years operation) don't have an optimized cloud strategy. So he provides a proposed three-stage model (preliminary assessment, migration's plan creation, implementation and maintenance) for the strategy of using cloud computing and migration to it.
(Subramanian and Seshasaayee, 2014)	They Suggest that required non-software components for setting up a cloud infrastructure consist of connectivity and bandwidth, land for data centers, security. They offer that several important factors should be considered and analyzed when migrating to the cloud.
(Alkhalil et al., 2014)	The model of decision-making process for cloud migration is provided based on Knowledge Base Decision Support System (KBDSS), which comprises Six main steps: business strategy, cloud environment, service suitability, risk assessment, vendor evaluation, and implementation.
(Jermyn et al., 2014)	They focus on the migration from the data center to cloud and propose a method that automatically computes optimized target resources and identifies required configurations. In this method, the relationships between servers are examined in the source environment. Then this method identifies potential resource consolidations and configuration. This information is applied to design a placement strategy for the target cloud architecture. This method reduces service's cost down to 60.1%.
(Pardeshi, 2014)	In order to migrate to system based cloud computing, a five-phase strategy, including preparation, analysis, and migration, concluding the cloud migration, maintenance and vendor management is presented.
(Kiadehi, 2014)	He offers a four-stage framework by reviewing the requirements and cloud computing strategies of the federal government, Department of Defense (DoD), Federal Aviation Administration (FAA) and the Government of Australia. Steps include decision-making, preparation, establishment, delivery and monitoring.
(Rockmann et al., 2014)	To implement cloud computing technology in an organization, three dimensions of required IT capabilities including technological, human, organizational capabilities are defined.
(Abderrahim and Choukair, 2014)	They develop Cloud Computing Adaptive Migration (CCAM) model By using Jamshidi's Cloud Reference migration model (RMM) and Enterprise Architecture Framework (EA). This model comprises five phases: baseline architecture, target architecture, migration decision, migration strategy and integration concerns. It is suggested divide baseline architecture into four domains of business, applications, data and infrastructure. They defin different scenarios of these four parts in order to migrate to the cloud. Reference Model (RM) or Reference Architecture (RA) should be determined before making a decision. Also during and after migration, activities continuity and interoperability of migrated elements with current elements in the organization and other providers should be ensured.
(Botto et al., 2014)	By reviewing papers from 2006 to 2013, 48 papers are selected to review migration of service oriented architecture (SOA) applications to cloud computing environments. Resources categorized in the areas of migration strategies (Conventional or model-driven), migration approach (re-host, re-factor, revise and re-build), migration type (replace a component, partially migrate, migrate the whole software stack and cloudify) and considered qualitative aspects.
(Mateescu et al., 2014)	They present a brief introduction of web-based tool to audit cloud computing migration which is called "Migration Assessment Tool (MAT)". This tool asks questions about the complexity of the implementation (challenges of migration to the cloud), risk and compliance, infrastructure, performance, etc. Then the MTA scores and computes the impact of the cloud adoption and suggests the suitable cloud service providers.
(Okai et al., 2014)	They identify security, privacy and reliability as the main challenges of adopting cloud computing and they set

References	Purpose of research and proposed output
	some guidelines for their proposed model in order to remove them. Also they provide a roadmap for migration, including planning stages, choosing the right deployment model, selecting the most suitable Service delivery model, vendor selection, negotiating the SLA and migration. The mentioned road map has been tested in the Asia Pacific University of Technology and Innovation [APU].
(Sefraoui et al., 2014)	They explain cloud type, deployment model, rationalization of resources and migration limitation for each type of organization.
(Garg et al., 2013)	They develop a framework (SMI Cloud) based on SMI (service measurement index). This framework helps customers to find the most appropriate cloud provider. The framework includes a service selection based on quality of service (QoS) requirements, service ranking based on the user's previous experience and performance of the service.
(Guillén, et al., 2013)	They proposed a service-oriented framework for developing cloud applications that allowed application construct from combination of software components and migrate freely between cloud platforms.
(Omerovic et al., 2013)	They provide a Decision Support Method (DSM) to select suitable cloud services and provider in multi-cloud environments, it that considers several aspects of cost, risk and quality. The DSM process consists of three main steps: establish context and model the target assess and verification of cost, risk, quality, and treatment and decision-making.
(Menychtas et al., 2013)	Based on ARTIST (advanced software-based service provisioning and migration of legacy software) methodology, a new approach is presented for modernization and migrate legacy application to the cloud environment, with taking into account the technical and economical aspects. ARTIST methodology includes pre-migration (technical and economic feasibility assessment, and application understanding), migration (using reverse engineering and forward engineering techniques), post- migration (verification of pre-defined goals), and evolution and reuse of migration's artifacts (maintenance and support activities of post-migration).
(Alonso et al., 2013)	By using ARTIST methodology, they present an approach consisting three main steps, the characterization of the legacy application from technical and business points of view, the technical feasibility analysis and economic feasibility analysis in order to decide migration to the cloud.
(Jamshidi et al., 2013)	With a systematic literature review of the migration, from 23 papers selected from the 2010 to 2013, they eventually offer Cloud-RMM (Cloud - Reference Migration Model). This model includes the processes of migration planning, migration execution, and migration evaluation; that the cross-cutting concerns cover these three processes in an umbrella form.
(Andrikopoulos et al., 2013a)	They define four types of migration (replacement, partially migrate, migrate the whole software stack and cloudify) as well as defining the adoption activities according to each type at different layers (data layer, business layer and the presentation layer).
(Andrikopoulos et al., 2013c)	They review challenges of migrating to cloud such as distribution of applications, multi-tenancy, QoS, cost of migration & operation, security and confidentiality of data, etc. finally they provide a decision support system.
(Andrikopoulos et al., 2013b)	With the aim of selecting an appropriate service provider for migration of existing applications based on least cost, they define a migration decision support system (MDSS) by analytical hierarchy process (AHP) that consists of three layers (user interface, offerings matcher and cost calculator).
(Chang et al., 2013)	They introduce a cloud computing business framework (CCBF) which consists of four main parts (classification, organizational sustainability modeling (OSM), service portability and linkage. This framework helps organizations in designing, deploying and migrating of cloud computing.
(Srirama et al., 2013)	They propose scientific computational experiments be done directly in the cloud and design a desktop to cloud migration (D2CM) tool to support direct migration of applications to the cloud. This tool supports life-cycle management for applications to be hosted on compatible infrastructure such as Amazon's EC2 or Eucalyptus.
(Menzel and Ranjan, 2012)	They divide the process of an IT system migration to a cloud infrastructure service into five steps and introduce a framework called cloud genius that offers a multi-criteria approach for deciding about the cloud IaaS service provider. The five steps are cloud infrastructure-service selection, selection of cloud VM image, cloud VM image customization, definition of migration strategy, and applying migration strategy.
(Johnson and Qu, 2012)	In order to decide cloud migration, they offer a holistic model of calculating income and cost of migration.
(Surendro and Fardani, 2012)	With the help of cloud computing reference model, they study the preparation of Indonesian companies related to the implementation of cloud computing. They said that most of companies tend to use public cloud and SaaS. There is not a problem about human resources, financial and infrastructure issues. The main problems are lack of managers' adequate training and lack of rules.
(Chauhan and Babar, 2012)	They propose a process framework to support migration that includes identification of requirements, identification of potential cloud hosting environments, analyzing application compatibility with potential cloud environments, identification of potential architecture solutions, and evaluation of cloud platforms, evaluation of potential architecture solutions, and implementation and system refactoring.
(Pfitzmann and Joukov, 2011)	In order to migrate to cloud in multi- image environments, they use a method that considers the analysis of return on investment (ROI) rate.
(Mircea and Andreescu, 2011)	They introduce processes of choosing and migrating to the cloud by using several tables: data identification, data classification, data evaluation and activity evaluation, and etc.
(Misra and Mondal, 2011)	They describe four key characteristics of the company's IT resources that should be considered during the migration. These are size, the utilization pattern, sensitivity of the data and criticality of work.
(Hajjat et al., 2011)	They contribute making decision for having a beneficial migration of applications to the cloud by modeling various factors such as: policy constraints, benefits, internet costs and increasing transaction delays.
(Khajeh-Hosseini et al., 2011)	In their study, two tools for cloud migration decisions are offered and besides usage of infrastructure as a service (IaaS). The first tool is applied for estimating the cost of using and the latter is a spreadsheet that classifies and evaluates the advantages and risks of using IaaS.
(Alabbadi, 2011)	For cloud formation, he introduces the cloud computing formation (C3F) model that belongs to the Jericho forum. Then he describes its dimensions such as physical location and management, ownership, and the architectural mindset. Next, they mapped IT activities of an educational organization to C3F model based on sensitivity and mission sensitivity matrix.

References	Purpose of research and proposed output
(Kundra, 2011)	He offers the decision framework of the federal government's cloud computing strategy in order to migrate services and applications to the cloud including: selection, preparation and management stages. He also offers "value" and "readiness" table to select applications and services for transmission.
(Garg et al., 2011)	They describes service measurement index (presented by the consortium of the cloud measurement index) for reviewing the quality of service level by AHP method and selection of appropriate services.
(Babar and Chauhan, 2011)	They describe their experiences of migrating an open source application called Hackstat to the cloud which is included the initial architecture of the application, and modified architecture according to the cloud requirements and decision plan.
(Greenwood et al., 2010)	They present, cloud adoption toolkit that addresses the challenges of cloud adoption with risk analysis and cost calculations.
(Li et al., 2010)	Public cloud providers are compared based on defined indexes in their study.
(Chan and Chieu, 2010)	They rank and map apps to cloud computing services by choosing the best cloud service provider.
(Khajeh-Hosseini et al., 2010b)	They investigate the advantages and risks of the migration of an Oil and Gas Company data center to Amazon EC2 and they show the cost of the infrastructure system within 5 years would be decreased 37% and support activities would be also decreased 21%.
(Lewis et al., 2005)	They study a smart technique that is a service-oriented migration and reuse technique in the field of migration. This technique helps organizations analyze their existing systems, whether their function can appear as a service in a service-oriented architecture.

Appendix B

Table B1
Concepts, sub-categories and categories as well as their extraction resources

No	Concept	References	Sub-category	Category	
1	Developing and expanding the knowledge about cloud computing	(Khan & Al-Yasiri, 2016),(Yaghmaei & Binesh, 2015),(Pardeshi, 2014),(Alkhalil, et al., 2014)	Expanding the knowledge about cloud computing and their basic requirements	Initiation phase	
2	Changing the mentality	(Kiadehi, 2014),(Rewatkar & Lanjewar, 2010),(Kundra, 2011)			
3	Understanding the legal requirement and implication related to the cloud.	(Alsufyani, et al., 2015)			
4	Become fully aware of the cloud computing principles	(Alkhalil, et al., 2014)			
5	Reviewing the standards related to the cloud.	(Khan & Al-Yasiri, 2016),(Kiadehi, 2014)			
6	Become fully aware of the cloud contract terms	(Khan & Al-Yasiri, 2016)			
7	Identifying the services and business processes.	(Yaghmaei & Binesh, 2015),(Rockmann, et al., 2014)	Identification of opportunities and threats		Initiation phase
8	Identifying the cloud environment.	(Alkhalil, et al., 2014)			
9	Identifying opportunities.	(Alkhalil, et al., 2014)			
10	Identifying limitations (constraints).	(Khan & Al-Yasiri, 2016),(Alkhalil, et al., 2014),(Alsufyani, et al., 2015)			
11	Identifying risks.	(Khan & Al-Yasiri, 2016),(Omerovic, et al., 2013),(Alkhalil, et al., 2014),(Kiadehi, 2014),(Khajeh-Hosseini, Greenwood, & Sommerville, 2010)			
12	Developing the knowledge base about cloud computing	(Andrikopoulos, Song, et al., 2013),(Alkhalil, et al., 2014)			
13	Understanding the applications.	(Menychtas, et al., 2013),(Khan & Al-Yasiri, 2016),(Rai, Sahoo, & Mehfuz, 2015)	Identification of detail requirements and solutions	Adoption phase	
14	Identifying the IT requirements.	(Yaghmaei & Binesh, 2015)			
15	Identifying the stakeholder requirement.	(Khan & Al-Yasiri, 2016),(Pardeshi, 2014),(Kiadehi, 2014),(Alkhalil, et al., 2014),(Chauhan & Babar, 2012)			
16	Identifying security requirements.	(Subramanian & Seshasaayee, 2014)			
17	Identifying compatibility requirements.	(Subramanian & Seshasaayee, 2014)			
18	Identifying the potential cloud hosting environment	(Chauhan & Babar, 2012)			
19	Identifying potential solutions.	(Chauhan & Babar, 2012),(Alkhalil, et al., 2014)			
20	Finding the optimized target resource.	(Jermyn, et al., 2014)			
21	Analyzing the cloud computing values (agility, efficiency, innovation, cost saving, availability)	(Kiadehi, 2014),(Kundra, 2011)	Preliminary Strategic analysis		Adoption phase
22	Evaluating enterprise readiness.	(Khan & Al-Yasiri, 2016),(Kiadehi, 2014)			
23	Evaluating governmental service specification	(Kiadehi, 2014)			
24	Evaluating the market landscape.	(Kiadehi, 2014)			
25	Analyzing the business strategy.	(Kiadehi, 2014)			
26	Evaluating the technology vision.	(Kiadehi, 2014)			
27	Risk assessment and ranking these risks (dependent on probability, affect, etc.)	(Khan & Al-Yasiri, 2016)			

No	Concept	References	Sub-category	Category
28	Reviewing the IT governance.	(Khan & Al-Yasiri, 2016)		
29	Evaluating the current stage of enterprise IT.	(Yaghmaei & Binesh, 2015),(Rai, Sahoo, & Mehfuz, 2015)		
30	SWOT analysis	(Pardeshi, 2014)		
31	Setting goals and target attribute (deployment, service and ownership models)	(Omerovic, et al., 2013),(Menychtas, et al., 2013)	Goal setting	Decision making and selection
32	Tuning the bench point for legal compatibility and security.	(Pardeshi, 2014)		
33	Specify the architecture of the target.	(Omerovic, et al., 2013)		
34	Characterizing the objectives aligned with enterprise strategy.	(Khan & Al-Yasiri, 2016)		
35	Analyzing stakeholder requirements.	(Jamshidi, et al., 2013),(Pardeshi, 2014)	Analyze the detail requirements and solutions	
36	Analyzing software and hardware requirements the cloud point of view.	(Pardeshi, 2014),(Khan & Al-Yasiri, 2016)		
37	Analyzing transparency, certificates and audits.	(Subramanian & Seshasaayee, 2014)		
38	Analyzing required it capabilities (technical, human, organizational).	(Rockmann, et al., 2014)		
39	Managerial and technical feasibility analysis.	(Pardeshi, 2014),(Khan & Al-Yasiri, 2016),(Jamshidi, et al., 2013),(Alonso, et al., 2013),(Menychtas, et al., 2013),(Kiadehi, 2014)		
40	Analyzing IT outsourcing capability.	(Rockmann, et al., 2014)		
41	Analyzing organizational attributes	(Rockmann, et al., 2014)		
42	Economical feasibility analysis	(Pardeshi, 2014),(Khan & Al-Yasiri, 2016),(Jamshidi, et al., 2013),(Alonso, et al., 2013),(Menychtas, et al., 2013),(Johnson & Qu, 2012),(Pfitzmann & Joukov, 2011),(Khajeh-Hosseini, Greenwood, & Sommerville, 2010)		
43	Analyzing infrastructure flexibility.	(Rockmann, et al., 2014)	Choosing the solutions	
44	Analyzing application compatibility with potential cloud environment.	(Kiadehi, 2014),(Chauhan & Babar, 2012)		
45	Developing a cloud knowledge based decision support system	(Alkhalil, et al., 2014)		
46	Choosing the suitable implementation platform.	(Kiadehi, 2014)		
47	Evaluating the suitability of cloud-based services and cloud-based solutions.	(Alkhalil, et al., 2014),(Garg, et al., 2013),(Chauhan & Babar, 2012),(Yaghmaei & Binesh, 2015)		
48	Ranking the services based on user experiences and service performance.	(Garg, et al., 2013)		
49	Choosing appropriate service and solution.	(Garg, et al., 2013),(Menzel & Ranjan, 2012),(Yaghmaei & Binesh, 2015),(Omerovic, et al., 2013)		
50	Choosing an appropriate cloud service provider.	(Khan & Al-Yasiri, 2016),(Alkhalil, et al., 2014),(Andrikopoulos, Strauch, et al., 2013),(Kiadehi, 2014),(Jamshidi, et al., 2013)	Develop the Migration strategy	
51	Policy making and planning for cloud migration.	(Kiadehi, 2014)		
52	Categorizing the current services and data.	(Khan & Al-Yasiri, 2016),(Menzel & Ranjan, 2012),(Jamshidi, et al., 2013)		
53	Choosing the immigrant services.	(Jamshidi, et al., 2013),(Pardeshi, 2014),(Kundra, 2011)		
54	Assigning responsibilities.	(Khan & Al-Yasiri, 2016),(Pardeshi, 2014)		
				Migration

No	Concept	References	Sub-category	Category	
55	Preparing the interoperability strategy.	(Abderrahim & Choukair, 2014)	Develop the Migration strategy (Continued)	Migration (Continued)	
56	Ensuring the convergence between enterprise architecture and cloud computing	(Abderrahim & Choukair, 2014)			
57	Preparing the elastic strategy.	(Andrikopoulos, Strauch, et al., 2013)			
58	Preparing the multi tenancy requirements.	(Andrikopoulos, Strauch, et al., 2013)			
59	Modernization and adaptation of legacy application in the cloud environment	(Menychtas, et al., 2013)			
60	Preparing the outsourcing strategy.	(Pardeshi, 2014)			
61	Developing a transfer schedule.	(Kundra, 2011),(Kiadehi, 2014)			
62	Considering the required turn over.	(Khan & Al-Yasiri, 2016)			
63	Effective contracting.	(Khan & Al-Yasiri, 2016),(Kundra, 2011),(Kiadehi, 2014),(Pardeshi, 2014)			
64	Preparing an exit strategy.	(Kiadehi, 2014)			
65	Applying the migrating strategy.	(Chauhan & Babar, 2012),(Menzel & Ranjan, 2012)	Pilot test and migration		
66	Choosing the pilot project.	(Pardeshi, 2014)			
67	Test the systems.	(Khan & Al-Yasiri, 2016), (Jamshidi, et al., 2013),(Rai, Sahoo, & Mehfuz, 2015)			
68	Extracting and migrating the data.	(Jamshidi, et al., 2013),(Pardeshi, 2014),(Rai, Sahoo, & Mehfuz, 2015)			
69	Seamlessly migrate to cloud computing services.	(Alkhalil, et al., 2014)			
70	Monitoring the application for its availability & performance throughout the migration process.	(Khan & Al-Yasiri, 2016)	Migration process control and evaluation	Adaptation and control	
71	Ensuring integration between new systems and older systems.	(Abderrahim & Choukair, 2014),(Kiadehi, 2014),(Kundra, 2011),(Pardeshi, 2014),(Jermyn, et al., 2014)			
72	Ensuring the service continuity management throughout the migration process.	(Abderrahim & Choukair, 2014)			
73	Monitoring and controlling the migration process.	(Pardeshi, 2014)			
74	Technical and commercial goals verification	(Menychtas, et al., 2013),(Garg, et al., 2013),(Jamshidi, et al., 2013)	Adaptation		
75	Creating a new set of it skills depending on the needs.	(Kiadehi, 2014),(Kundra, 2011),(Rockmann, et al., 2014)			
76	Applying an agile approach to innovation in services.	(Kiadehi, 2014)			
77	IT governance alignment to organizational strategy	(Kiadehi, 2014)			
78	Architecture recovery and adaptation.	(Jamshidi, et al., 2013)			
79	Implementing a security structure in order to defend against cyber threats.	(Kiadehi, 2014)	Support and Maintenance	Routinization and maintenance	
80	Developing guidelines to guide migration.	(Kiadehi, 2014),(Pardeshi, 2014),(Omerovic, et al., 2013)			
81	Vendor management.	(Yaghmaei & Binesh, 2015),(Pardeshi, 2014)			
82	Updating and maintenance.	(Khan & Al-Yasiri, 2016),(Menychtas, et al., 2013),(Rai, Sahoo, & Mehfuz, 2015)			
83	Support.	(Menychtas, et al., 2013)			
84	Reviewing all related policies to ensure support for cloud computing services.	(Kiadehi, 2014)			

No	Concept	References	Sub-category	Category
85	Assigning a routine budget.	(Pardeshi, 2014)	Continual monitoring	Routinization and maintenance (Continued)
86	Monitoring the SLA.	(Kundra, 2011),(Garg, Toosi, Gopalaiyengar, & Buyya, 2014),(Kiadehi, 2014)		
87	Evaluating user quality of experiences.	(Pardeshi, 2014)		
88	Monitoring quality of services.	(Pardeshi, 2014),(Khan & Al-Yasiri, 2016)		
89	Ensuring the transparency, maintenance and updating.	(Khan & Al-Yasiri, 2016)		
90	Evaluating periodical service models.	(Kundra, 2011),(Kiadehi, 2014)		
91	Evaluating periodical vendors.	(Kundra, 2011),(Kiadehi, 2014)		

Appendix C

Table C1

Generic goals and practices of cloud migration process areas.

No.	Abb.	objective	General practices
1	GG1	Achieve specific goals	GP 1.1: Perform Specific Practices.
2	GG2	Institutionalize a managed process	GP 2.1: Establish an organizational policy GP 2.2: Plan the process GP 2.3: Provide resources GP 2.4: Assign responsibility GP 2.5: Train people GP 2.6: Manage configurations GP 2.7: Identify and involve relevant stakeholders GP 2.8: Monitor and control the process GP 2.9: Objectively evaluate adherence GP 2.10: Review status with higher level management
3	GG3	Institutionalize a defined process	GP 3.1: Establish a defined process GP 3.2: Collect improvement information

Appendix D

Table D1

Specific goals of cloud migration process areas.

No.	Process	Specific goals
1	Expanding the cloud knowledge	Developing and expanding the knowledge about cloud computing
2		Changing the mentality
3		Understanding the legal requirement and implication related to the cloud.
4		Become fully aware of the cloud computing principles
5		Reviewing the standards related to the cloud.
6		Become fully aware of the cloud contract terms
7	Identification of opportunities and threats	Identifying the services and business processes.
8		Identifying the cloud environment.
9		Identifying opportunities
10		Identifying limitations

No.	Process	Specific goals
11		Identifying risks.
12		Developing the knowledge base about cloud computing
13	Identification of detail requirements and solutions	Understanding the applications.
14		Identifying the IT requirements.
15		Identifying the stakeholder's requirement.
16		Identifying security requirements.
17		Identifying compatibility requirements.
18		Identifying the potential cloud hosting environment
19		Identifying potential solutions.
20		Finding the optimized target resource.
21	Preliminary strategic analysis	Analyzing the cloud computing values (agility, efficiency, innovation, cost saving, availability)
22		Evaluating enterprise readiness.
24		Evaluating the market landscape.
25		Analyzing the business strategy.
26		Evaluating the technology vision.
27		Risk assessment and ranking these risks (dependent on probability, affect, etc.)
28		Reviewing the IT governance.
29		Evaluating the current stage of enterprise IT.
30		SWOT analysis
31	Goal setting	Setting goals (technical and financial) and target attribute (deployment, service & ownership models)
32		Tuning the bench point for legal compatibility and security.
33		Specify the architecture of the target.
34		Characterizing the objectives aligned with enterprise strategy.
35	Analyze the detail requirements and solutions	Analyzing stakeholder requirements.
36		Analyzing software and hardware requirements the cloud point of view.
37		Analyzing transparency, certificates and audits.
38		Analyzing required it capabilities (technical, human, organizational).
39		Managerial & technical feasibility analysis.
40		Analyzing IT outsourcing capability.
41		Analyzing organizational attributes

No.	Process	Specific goals
42		Economical feasibility analysis
43		Analyzing infrastructure flexibility.
44		Analyzing application compatibility with potential cloud environment.
45	Choosing the solutions	Developing a cloud knowledge based decision support system
46		Choosing the suitable implementation platform.
47		Evaluating the suitability of cloud-based services and cloud-based solutions.
48		Ranking the services based on user experiences and service performance.
49		Choosing appropriate service and solution.
50		Choosing an appropriate cloud service provider.
51	Develop the migration strategy	Policy making and planning for cloud migration.
52		Categorizing the current services & data.
53		Choosing the immigrant services.
54		Assigning responsibilities.
55		Preparing the interoperability strategy.
56		Ensuring the convergence between enterprise architecture and cloud computing
57		Preparing the elastic strategy.
58		Preparing the multi tenancy requirements.
59		Modernization and adaptation of legacy application in the cloud environment
60		Preparing the outsourcing strategy.
61		Developing a transfer schedule.
62		Considering the required turn over.
63		Effective contracting.
64		Preparing an exit strategy.
65	Pilot test and migration	Applying the migrating strategy.
66		Choosing the pilot project.
67		Test the systems.
68		Extracting and migrating the data.
69		Seamlessly migrate to cloud computing services.
70	Migration process control and evaluation	Monitoring the application for its availability & performance throughout the migration process.
71		Ensuring integration between new systems and older systems.

No.	Process	Specific goals
72		Ensuring the service continuity management throughout the migration process.
73		Monitoring and controlling the migration process.
74		Technical and commercial goals verification
75	Adaptation	Creating a new set of IT skills depending on the needs.
76		Applying an agile approach to innovation in services.
77		IT governance alignment to organizational strategy
78		Architecture recovery and adaptation.
79	Support and maintenance	Implementing a security structure in order to defend against cyber threats.
80		Developing guidelines to guide migration.
81		Vendor management
82		Updating and maintenance
83		Support
84		Reviewing all related policies to ensure support for cloud computing services.
85		Assigning a routine budget.
86	Continual monitoring	Monitoring the SLA.
87		Evaluating user quality of experiences.
88		Monitoring quality of services.
89		Ensuring the transparency, maintenance and updating.
90		Evaluating periodical service models.
91		Evaluating periodical vendors.
92		Establish and maintain process-performance models for the migration processes.
93		The migration process is quantitatively managed
94	Optimizing	Improvements are proactively identified, evaluated using statistical and other quantitative techniques.
95		Measurable improvements to the organization's processes and technologies are deployed and evaluated using statistical and other quantitative techniques.
96	Infusion	Systematically determining the root causes of outcomes.
97		Systematically addressing the root causes of outcomes.
98		Ensuring the maximum usage of application's potential in a deep, broad and comprehensive way in the organizational or individual work system.

7. References

- Abderrahim, W., Choukair, Z., 2014. A framework architecture based model for cloud computing adaptive migration, 2014 Global Information Infrastructure and Networking Symposium, GIIS 2014, pp. 1 - 6.
- Adrees, S.M., Omer, K.a.M., Sheta, E.O., 2015. cloud computing architecture for higher education in the third world countries (republic of the sudan as model). *International Journal of Database Management Systems* 7, 13-24.
- Alabbadi, M.M., 2011. Cloud computing for education and learning: Education and learning as a service (ELaaS), *Interactive Collaborative Learning (ICL)*, 2011 14th International Conference on. IEEE, pp. 589-594.
- Alkhalil, A., Sahandi, R., John, D., 2014. Migration to Cloud Computing: A Decision Process Model, *Central European Conference on Information and Intelligent Systems*. Faculty of Organization and Informatics Varazdin, pp. 154-163.
- Alonso, J.M., Orue-Echevarria, L., Escalante, M., Gorroñoigoitia, J., Presenza, D., 2013. Cloud modernization assessment framework: Analyzing the impact of a potential migration to Cloud, *Maintenance and Evolution of Service-Oriented and Cloud-Based Systems (MESOCA)*, 2013 IEEE 7th International Symposium on the. IEEE, pp. 64-73.
- Alsaeed, N., Saleh, M., 2015. Towards Cloud Computing Services for Higher Educational Institutions: Concepts & Literature Review, *Cloud Computing (ICCC)*, 2015 International Conference on. IEEE, pp. 1-7.
- Alsaffar, A.A., Shin, Y.R., Huh, E.N., 2015. IPTV Service Framework Based on Secure Authentication and Lightweight Content Encryption for Screen-Migration in Cloud Computing. *Advances in Multimedia* 2015, 9-22.
- Alsufyani, R., Safdari, F., Chang, V., 2015. Migration of cloud services and deliveries to higher education, *Proceedings of ESaaS 2015-2nd International Workshop on Emerging Software as a Service and Analytics*, In conjunction with the 5th International Conference on Cloud Computing and Services Science-CLOSER 2015, pp. 86-94.
- Andrikopoulos, V., Binz, T., Leymann, F., Strauch, S., 2013a. How to adapt applications for the Cloud environment. *Computing* 95, 493-535.
- Andrikopoulos, V., Song, Z., Leymann, F., 2013b. Supporting the migration of applications to the cloud through a decision support system, 2013 IEEE Sixth International Conference on Cloud Computing. IEEE, pp. 565-572.
- Andrikopoulos, V., Strauch, S., Leymann, F., 2013c. Decision Support for Application Migration to the Cloud: Challenges and Vision *Proceedings of CLOSER'13*, 149-155.
- Babar, M.A., Chauhan, M.A., 2011. A tale of migration to cloud computing for sharing experiences and observations, *Proceedings - International Conference on Software Engineering*, Hawaii, USA, pp. 50-56.
- Botto, M., González-Huerta, J., Insfran, E., 2014. Are model-driven techniques used as a means to migrate SOA applications to cloud computing? *WEBIST 2014 - Proceedings of the 10th International Conference on Web Information Systems and Technologies* 1, 208-213.
- Chan, H., Chieu, T., 2010. Ranking and mapping of applications to cloud computing services by SVD, *Network Operations and Management Symposium Workshops (NOMS Wksp)*, 2010 IEEE/IFIP. IEEE, pp. 362-369.
- Chang, V., Walters, R.J., Wills, G., 2013. The development that leads to the Cloud Computing Business Framework. *International Journal of Information Management* 33, 524-538.
- Chauhan, M.A., Babar, M.A., 2012. Towards process support for migrating applications to cloud computing, *Cloud and Service Computing (CSC)*, 2012 International Conference on. IEEE, pp. 80-87.
- Cooper, R.B., Zmud, R.W., 1990. Information technology implementation research: a technological diffusion approach. *Management science* 36, 123-139.
- Davis, F.D., 1989. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS quarterly*, 319-340.
- Douglas, A.C., Mills, J.E., Niang, M., Stepchenkova, S., Byun, S., Ruffini, C., Lee, S.K., Loutfi, J., Lee, J.-K., Atallah, M., 2008. Internet addiction: Meta-synthesis of qualitative research for the decade 1996–2006. *Computers in Human Behavior* 24, 3027-3044.
- El-Gazzar, R., Hustad, E., Olsen, D.H., 2016. Understanding cloud computing adoption issues: A Delphi study approach. *Journal of Systems and Software* 118, 64-84.
- Gangwar, H., Date, H., Ramaswamy, R., 2015. Developing a Cloud-Computing Adoption Framework. *Global Business Review* 16, 632-651.
- García-Galán, J., Trinidad, P., Rana, O.F., Ruiz-Cortés, A., 2016. Automated configuration support for infrastructure migration to the cloud. *Future Generation Computer Systems* 55, 200-212.
- Garg, S.K., Toosi, A.N., Gopalaiyengar, S.K., Buyya, R., 2014. SLA-based virtual machine management for heterogeneous workloads in a cloud datacenter. *Journal of Network and Computer Applications* 45, 108-120.
- Garg, S.K., Versteeg, S., Buyya, R., 2011. Smicloud: A framework for comparing and ranking cloud services, *Utility and Cloud Computing (UCC)*, 2011 Fourth IEEE International Conference on. IEEE, Melbourne, Australia, pp. 210-218.
- Garg, S.K., Versteeg, S., Buyya, R., 2013. A framework for ranking of cloud computing services. *Future Generation Computer Systems* 29, 1012-1023.
- Gholami, M.F., Daneshgar, F., Low, G., Beydoun, G., 2016. Cloud migration process—A survey, evaluation framework, and open challenges. *Journal of Systems and Software* 120, 31-69.
- Greenwood, D., Khajeh-Hosseini, A., Smith, J., Sommerville, I., 2010. The cloud adoption toolkit: Addressing the challenges of cloud adoption in enterprise. *Arxiv preprint*.
- Guillén, J., Miranda, J., Murillo, J.M., Canal, C., 2013. A service-oriented framework for developing cross cloud migratable software. *Journal of Systems and Software* 86, 2294-2308.
- Hajjat, M., Sun, X., Sung, Y.-W.E., Maltz, D., Rao, S., Sripanidkulchai, K., Tawarmalani, M., 2011. Cloudward bound: planning for beneficial migration of enterprise applications to the cloud. *ACM SIGCOMM Computer Communication Review* 41, 243-254.
- Hazen, B.T., Overstreet, R.E., Cegielski, C.G., 2012. Supply chain innovation diffusion: going beyond adoption. *The International Journal of Logistics Management* 23, 119-134.
- Hwang, J., Bai, K., Tacci, M., Vukovic, M., Anerousis, N., 2016. Automation and orchestration framework for large-scale enterprise cloud migration. *IBM Journal of Research and Development* 60, 1: 1-1: 12.

- Jamshidi, P., Ahmad, A., Pahl, C., 2013. Cloud migration research: a systematic review. *Cloud Computing, IEEE Transactions on* 1, 142-157.
- Jasperson, J.S., Carter, P.E., Zmud, R.W., 2005. A comprehensive conceptualization of post-adoptive behaviors associated with information technology enabled work systems. *Mis Quarterly* 29, 525-557.
- Jermyn, J., Hwang, J., Bai, K., Vukovic, M., Anerousis, N., Stolfo, S., 2014. Improving readiness for enterprise migration to the cloud, *Proceedings of the Middleware Industry Track. ACM, NY, USA*, 2014, pp. 51-57.
- Johnson, B., Qu, Y., 2012. A holistic model for making cloud migration decision: A consideration of security, architecture and business economics, *Parallel and Distributed Processing with Applications (ISPA), 2012 IEEE 10th International Symposium on. IEEE*, pp. 435-441.
- Khajeh-Hosseini, A., Greenwood, D., Smith, J.W., Sommerville, I., 2010a. The cloud adoption toolkit: Addressing the challenges of cloud adoption in enterprise. *arXiv preprint arXiv:1003.3866*.
- Khajeh-Hosseini, A., Greenwood, D., Sommerville, I., 2010b. Cloud migration: a case study of migrating an enterprise IT system to IaaS, *Cloud Computing (CLOUD), 2010 IEEE 3rd International Conference on. IEEE, Miami*, 2010, pp. 450-457.
- Khajeh-Hosseini, A., Sommerville, I., Bogaerts, J., Teregowda, P., 2011. Decision support tools for cloud migration in the enterprise, *Cloud Computing (CLOUD), 2011 IEEE International Conference on. IEEE*, pp. 541-548.
- Khan, N., Al-Yasiri, A., 2016. Framework for cloud computing adoption: A road map for Smes to cloud migration. *International Journal on Cloud Computing: Services and Architecture (IJCCSA)* 5, 1-15.
- Kiadehi, E.M., S, 2014. Cloud Computing Technology in Iran: Opportunities, Threats. *International Journal of Electronics Communication and Computer Engineering* 5, 166-172.
- Kumar, A., Mishra, S., Mishra, A., 2015. Priority with adoptive data migration in case of disaster using cloud computing use style, *Proceedings - 2015 International Conference on Communication, Information and Computing Technology, ICCICT 2015, Mumbai, India*, pp. 1-6.
- Kundra, V., 2011. Federal cloud computing strategy. Washington D.C: The White House.
- Kwon, T.H., Zmud, R.W., 1987. Unifying the fragmented models of information systems implementation, *Critical issues in information systems research. John Wiley & Sons, Inc., Chichester, UK*, 1987, pp. 227-251.
- Lewis, G., Morris, E., Smith, D., 2005. Service-oriented migration and reuse technique (smart), *Software Technology and Engineering Practice*, 2005. 13th IEEE International Workshop on. IEEE, pp. 15-23.
- Li, A., Yang, X., Kandula, S., Zhang, M., 2010. CloudCmp: comparing public cloud providers, *Proceedings of the 10th ACM SIGCOMM conference on Internet measurement. ACM, Melbourne, Australia*, pp. 1-14.
- Markovic, D.S., Zivkovic, D., Branovic, I., Popovic, R., Cvetkovic, D., 2013. Smart power grid and cloud computing. *Renewable and Sustainable Energy Reviews* 24, 566-577.
- Mateescu, G., Vladescu, M., Sgarciu, V., 2014. Auditing cloud computing migration, *SACI 2014 - 9th IEEE International Symposium on Applied Computational Intelligence and Informatics, Proceedings, Timisoara*, pp. 263-268.
- McClean, S., Shaw, A., 2005. From schism to continuum? The problematic relationship between expert and lay knowledge—an exploratory conceptual synthesis of two qualitative studies. *Qualitative Health Research* 15, 729-749.
- Mell, P., Grance, T., 2011. The NIST definition of cloud computing. National Institute of Standards and Technology. Special Publication 800-145.
- Menychtas, A., Santzaridou, C., Kousiouris, G., Varvarigou, T., Orue-Echevarria, L., Alonso, J.M., Gorronogoitia, J., Bruneliere, H., Strauss, O., Senkova, T., 2013. ARTIST Methodology and Framework: A novel approach for the migration of legacy software on the Cloud, *Symbolic and Numeric Algorithms for Scientific Computing (SYNASC), 2013 15th International Symposium on. IEEE*, pp. 424-431.
- Menzel, M., Ranjan, R., 2012. CloudGenius: decision support for web server cloud migration, *Proceedings of the 21st international conference on World Wide Web. ACM, Lyon, France*, pp. 979-988.
- Mircea, M., Andresescu, A.I., 2011. Using cloud computing in higher education: A strategy to improve agility in the current financial crisis. *Communications of the IBIMA* 2011, 1-15.
- Misra, S.C., Mondal, A., 2011. Identification of a company's suitability for the adoption of cloud computing and modelling its corresponding Return on Investment. *Mathematical and Computer Modelling* 53, 504-521.
- Oates, B.J., 2006. *Researching information systems and computing*, 1 ed. Sage Publications, London.
- Okai, S., Uddin, M., Arshad, A., Alsaqour, R., Shah, A., 2014. Cloud computing adoption model for universities to increase ICT proficiency. *SAGE Open* 4, 1-10.
- Omerovic, A., Muntés-Mulero, V., Matthews, P., Gunka, A., 2013. Towards a method for decision support in multi-cloud environments. *CLOUD COMPUTING* 2013, 244-250.
- Pardeshi, V.H., 2014. Cloud Computing for Higher Education Institutes: Architecture, Strategy and Recommendations for Effective Adaptation. *Procedia Economics and Finance* 11, 589-599.
- Pfitzmann, B., Joukov, N., 2011. Migration to multi-image cloud templates, *Services Computing (SCC), 2011 IEEE International Conference on. IEEE*, pp. 80-87.
- Phaphoom, N., Wang, X., Samuel, S., Helmer, S., Abrahamsson, P., 2015. A survey study on major technical barriers affecting the decision to adopt cloud services. *Journal of Systems and Software* 103, 167-181.
- Prasad, A., Green, P., Heales, J., Finau, G., 2014. On cloud computing service considerations for the small and medium enterprises, *Twentieth Americas Conference on Information Systems*.
- Premkumar, G., 2003. A meta-analysis of research on information technology implementation in small business. *Journal of organizational computing and electronic commerce* 13, 91-121.
- Rai, R., Sahoo, G., Mehruz, S., 2015. Exploring the factors influencing the cloud computing adoption: a systematic study on cloud migration. *SpringerPlus* 4, 1-12.
- Rewatkar, L.R., Lanjewar, U., 2010. Implementation of cloud computing on web application. *International Journal of Computer Applications* 2, 28-32.

- Rimal, B.P., Choi, E., Lumb, I., 2009. A taxonomy and survey of cloud computing systems, INC, IMS and IDC, 2009. NCM'09. Fifth International Joint Conference on. Ieee, pp. 44-51.
- Rockmann, R., Weeger, A., Gewald, H., 2014. Identifying Organizational Capabilities for the Enterprise-Wide Usage of Cloud Computing, PACIS, p. 355.
- Sabi, H.M., Uzoka, F.M.E., Langmia, K., Njeh, F.N., 2016. Conceptualizing a model for adoption of cloud computing in education. *International Journal of Information Management* 36, 183-191.
- Safari, F., Safari, N., Hasanzadeh, A., 2015a. The adoption of software-as-a-service (SaaS): ranking the determinants. *Journal of Enterprise Information Management* 28, 400-422.
- Safari, N., Safari, F., Kazemi, M., Ahmadi, S., Hasanzadeh, A., 2015b. Prioritisation of cloud computing acceptance indicators using fuzzy AHP. *International Journal of Business Information Systems* 19, 488-504.
- Saga, V.L., Zmud, R.W., 1993. The nature and determinants of IT acceptance, routinization, and infusion, *Proceedings of the IFIP TC8 working conference on diffusion, transfer and implementation of information technology*. Elsevier Science Inc., pp. 67-86.
- Sandelowski, M., Barroso, J., 2007. *Handbook for synthesizing qualitative research*. Springer Publishing Company, New York.
- Sefraoui, O., Aissaoui, M., Eleuldj, M., 2014. Cloud computing migration and IT resources rationalization, *International Conference on Multimedia Computing and Systems -Proceedings*, pp. 1164-1168.
- Sharma, T., Banga, V.K., 2013. Efficient and Enhanced Algorithm in Cloud Computing. *International Journal of Soft Computing and Engineering* 13, 2231-2307.
- Siclovan, A., 2012. The Future of Cloud Computing. *BRAND. Broad Research in Accounting, Negotiation, and Distribution* 3, pp. 36-39.
- Srirama, S.N., Ivanistsev, V., Jakovits, P., Willmore, C., 2013. Direct migration of scientific computing experiments to the cloud, *Proceedings of the 2013 International Conference on High Performance Computing and Simulation, HPCS 2013*, pp. 27-34.
- Subramanian, S., Seshasaayee, A., 2014. Review & Proposal for a Cloud based Framework for Indian Higher Education. *International Journal Of Engineering And Computer Science* 3.
- Surendro, K., Fardani, A., 2012. Identification of SME readiness to implement cloud computing, *Cloud Computing and Social Networking (ICCCSN)*, pp. 1-4.
- Tashkandi, A., Al-Jabri, I., 2015. Cloud Computing Adoption by Higher Education Institutions in Saudi Arabia: Analysis Based on TOE, 2015 International Conference on Cloud Computing, ICCC 2015, Riyadh, pp. 1-8.
- Tayal, S., 2011. Tasks scheduling optimization for the cloud computing systems. *International Journal of Advanced Engineering Sciences And Technologies (IAEST)* 5, 111-115.
- Team, C.P., 2011. *CMMI for Services Version 1.3*. Carnegie Mellon University publication. , Pittsburgh, Pennsylvania.
- Thorne, S., 2009. The role of qualitative research within an evidence-based context: can metasynthesis be the answer? *International journal of nursing studies* 46, 569-575.
- Tornatzky, L., Fleischer, M., 1990. *The Processes of Technological Innovation*. Lexington, MA: Lexington Books.
- Torreçilla-Salinas, C., Sedeño, J., Escalona, M., Mejías, M., 2016. Agile, Web Engineering and Capability Maturity Model Integration: A systematic literature review. *Information and Software Technology* 71, 92-107.
- Venkatesh, V., Davis, F.D., 2000. A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management science* 46, 186-204.
- Venkatesh, V., Morris, M.G., Davis, G.B., Davis, F.D., 2003. User acceptance of information technology: Toward a unified view. *MIS quarterly*, 425-478.
- Venkatesh, V., Thong, J.Y., Xu, X., 2012. Consumer acceptance and use of information technology: extending the unified theory of acceptance and use of technology. *MIS quarterly* 36, 157-178.
- Wielki, J., 2015. An analysis of the opportunities and challenges connected with utilization of the cloud computing model and the most important aspects of the migration strategy, *Proceedings of the 2015 Federated Conference on Computer Science and Information Systems, FedCSIS 2015*, pp. 1569-1574.
- Yaghmaei, O., Binesh, F., 2015. Impact of Applying Cloud Computing On Universities Expenses. *IOSR Journal of Business and Management* 17, 42-47.
- Zimmer, L., 2006. Qualitative meta-synthesis: a question of dialoguing with texts. *Journal of advanced nursing* 53, 311-318.
- Zubrow, D., 2003. Current trends in adoption of the CMMI® Product Suite, *Computer Software and Applications Conference, 2003. COMPSAC 2003. Proceedings. 27th Annual International. IEEE, Dallas, TX, USA*, pp. 126-129.

Hamid reza Bazi is currently the Ph.D. candidate in information technology management at the Tarbiat Modares University. His major research interests focus on cloud computing and migration models.

Alireza Hasanzadeh is an academic who graduated from the University of Tehran with a Ph.D. in systems management. He is an associate Professor & Head of Information Technology Management Department at the Tarbiat Modares University. He has published many works in IT-IS subjects in international journal of information management, Knowledge-Based Systems, Computers & Education, Expert Systems with Applications. He has a research interest in analysis and design and implementation of integrated management information systems, intelligent information systems, cloud computing, and service oriented architecture.

Ali Moeini received his Ph.D. in Nonlinear Systems at the University of Sussex, UK, in 1997. He is an Associate Professor of Electrical & Computer Engineering at Faculty of Engineering Science, School of Engineering, the University of Tehran, and ad joint Professor at Department of Information Technology Management, Faculty of Management, the University of Tehran in Iran. His research interests include Formal Methods in Knowledge and Software Engineering and Knowledge Management, Soft computing Methods, Randomized Algorithms, Approximation Algorithms, and Bioinformatics. He has published many papers in Journals and Conferences on the above mentioned topics.